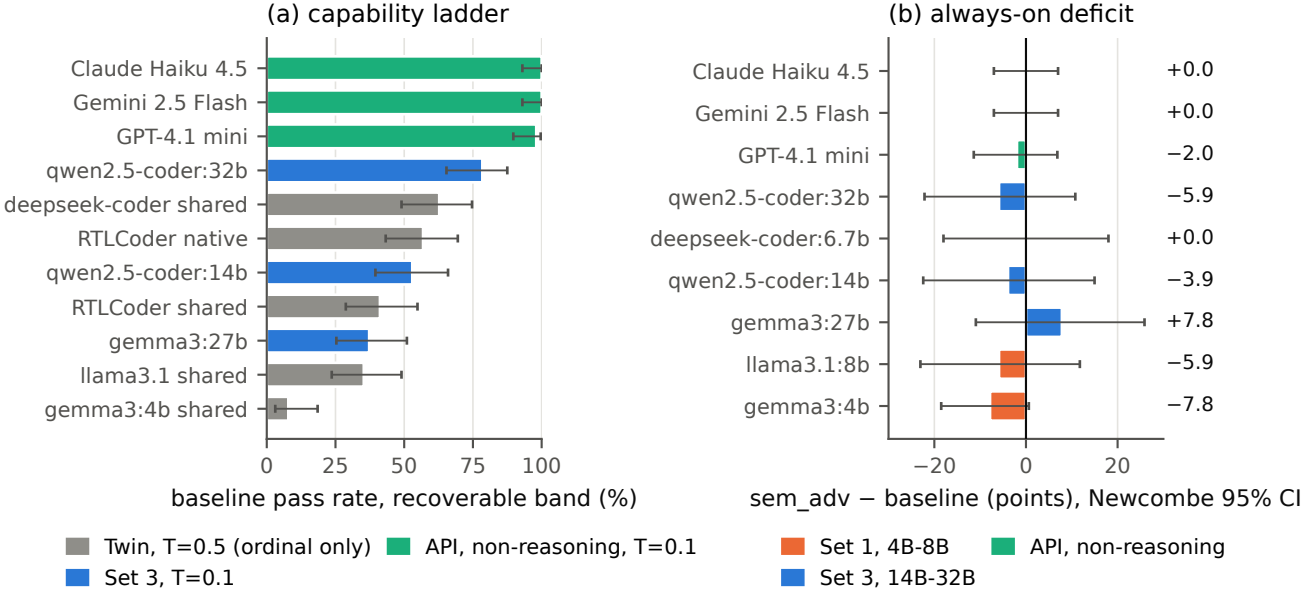
(a) capability ladder
Claude Haiku 4.5
Gemini 2.5 Flash
GPT-4.1 mini
qwen2.5-coder:32b
deepseek-coder shared
RTLCoder native
qwen2.5-coder:14b
RTLCoder shared
gemma3:27b
llama3.1 shared
gemma3:4b shared
0
25
50
75
100
baseline pass rate, recoverable band (%)
(b) always-on deficit
Claude Haiku 4.5
Gemini 2.5 Flash
GPT-4.1 mini
qwen2.5-coder:32b
deepseek-coder:6.7b
qwen2.5-coder:14b
gemma3:27b
llama3.1:8b
gemma3:4b
+0.0
+0.0
−2.0
−5.9
+0.0
−3.9
+7.8
−5.9
−7.8
−20
0
20
sem_adv − baseline (points), Newcombe 95% CI
Twin, T=0.5 (ordinal only)
API, non-reasoning, T=0.1
Set 3, T=0.1
Set 1, 4B-8B
API, non-reasoning
Set 3, 14B-32B